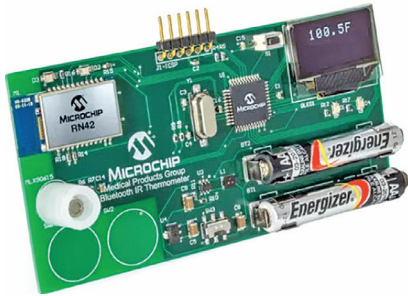


Thermometer With Capacitive Touch And Bluetooth

The connected thermometer board combines the MCU's low-power efficiency and touch sensitivity with the Bluetooth module's connectivity, offering a versatile digital thermometer solution with advanced features like a 28-channel ADC and global certification for seamless integration into wireless systems.

Microchip's connected thermometer board demonstrates the implementation of a Bluetooth-connected digital thermometer using Microchip's PIC16F1519 MCU and RN42 Bluetooth module. The MCU is an eXtreme Low Power (XLP), cost-effective 8-bit microcontroller with an integrated capacitive touch sensing module. Connectivity is demonstrated on this board using Microchip's fully qualified RN42 Bluetooth 2.1 + EDR module. It is designed with eXtreme Low Power (XLP) technology for efficient power management and includes an mTouch capacitive touch sensing module for responsive interfaces. It also features a 28-channel 10-bit analogue-to-digital converter (ADC) for accurate signal conversion and is equipped with a fixed voltage reference with selectable outputs for added versatility. Available in multiple packages, such as the compact ultra-thin QFN (UQFN), it meets various application needs. The RN4871 Bluetooth Low Energy module, certified with Bluetooth 5, enhances connectivity, and includes an ASCII command interface for easy communication. Certified globally, including in the USA, Canada, Brazil, Europe, and Asia, it supports mobile applications via Microchip Bluetooth Data (MBD) and offers remote configuration. It allows the choice of a ceramic chip antenna or an external antenna, providing design flexibility in wireless systems.

These designs can be used in various healthcare sectors.

Microchip

<https://www.electronicsforu.com/electronics-projects/thermometer-reference-design-with-integrated-mcu-capacitive-touch-module-and-bluetooth>

Skin Temperature Thermometer

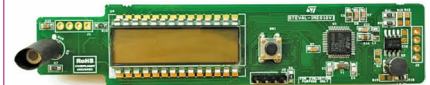
The reference design integrates a high-accuracy TMP117 sensor and CC2640R2F BLE MCU for seamless temperature data transmission to Bluetooth devices, offering precise ($\pm 0.1^{\circ}\text{C}$) measurements and versatile applications in healthcare and wearable tech.

The Texas Instruments (TI) TIDA-01624 reference design features the TMP117 high-accuracy, digital temperature sensor for direct skin contact, coupled with the CC2640R2F SimpleLink Bluetooth Low Energy (BLE) microcontroller (MCU) for data transmission. This setup allows 16-bit digital temperature data to be sent via I²C to the CC2640R2F, which then communicates this information to a Bluetooth-enabled device. Designed for efficiency, the patch operates on a 3V, flexible thin-film battery, but can also be powered externally through large contact pads for testing purposes when the battery is not connected. For temperature monitoring, an alternative approach involves a removable cover, akin to probe-type thermometers used for oral measurements, requiring system and cover characterisation in the final design. Probe-based systems might allow mode switching through software adjustments, necessitating an MCU with Bluetooth capabilities for the probe. This design finds application across medical devices, healthcare, and wearable technology, offering tailored solutions to enhance functionality and efficiency in these sectors. It features high accuracy ($\pm 0.1^{\circ}\text{C}$) for human body temperature measurements, a 2.4GHz RF transceiver compatible with BLE 4.2 and 5.

They apply in fields including medical devices, healthcare, and wearable technology.

Texas Instruments (TI)

<https://www.electronicsforu.com/electronics-projects/skin-temperature-measurement-reference-design>

Infrared Forehead Thermometer

The reference design for infrared forehead thermometer makes checking temperature easy and accurate without touching the skin, uses little battery, and lets you track temperatures and connect with an app for easy use.

The STEVAL-IME010V1 evaluation board, an advanced infrared forehead thermometer by STMicroelectronics, utilises the STM8L052C6 microcontroller along with the M24LR04E-R 4kb Dynamic NFC/RFID tag and an infrared sensor to offer accurate human body temperature measurements. Unlike traditional thermometers, it provides gentle, non-invasive readings with high precision, designed for a wide environmental temperature range from -40°C to $+80^{\circ}\text{C}$. It accurately measures body temperatures within 20°C to 42.2°C , with a minimal margin of error of $\pm 0.5^{\circ}\text{C}$. The device enhances convenience with a clear LCD panel that shows important information, including temperature in Celsius or Fahrenheit according to user preference. Its design supports IR sensors MLX90614 and MLX90615 on one PCB, improving usability and featuring a low battery indicator to keep the device running when needed. It can store 60 temperature readings for detailed tracking and analysis, while an Android app facilitates data logging via RF EEPROM and NFC connection. Energy efficiency is a key aspect, as the design consumes only 1.2 microamperes in sleep mode, extending battery life and adhering to RoHS standards for environmental sustainability. This innovation represents a leap in non-invasive temperature measurement, combining precision, adaptability, and effectiveness.

The reference design is suitable for healthcare and medical facilities.

STMicroelectronics

<https://www.electronicsforu.com/electronics-projects/reference-design-for-infrared-forehead-thermometer>